

DATABASE DESCRIPTION “SMLBase”

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The SMLBase is a database of box models that describe mixing and burial of sedimentary particles at specific sampling sites. Each box model is described by a sedimentation rate, one or two bioturbation depths, and one or two biodiffusion coefficients.

The database is given as a .csv file named “SMLBase.csv”. It contains the following fields:

- boxModelID: Inique ID assigned to each box model
- dataContributors: Short reference of all publications that contributed to the box model. See “References” for the full bibliographic information
- coreName: Name of the core/sampling location in the original publication
- S: Sedimentation rate (in cm/a)
- SMin: Lower bound on the sedimentation rate (in cm/a)
- SMax: Upper bound on the sedimentation rate (in cm/a)
- STracer: Tracer used to estimate sedimentation rate
- SComment: Comments on the entries about sedimentation rate
- M1: Mixing intensity from the sediment surface to L1 (in cm^2/a)
- M2: Mixing intensity from L1 to L2 . If no value is given, the model is a one box model (in cm^2/a)*
- M1Min: Lower bound on M1 (in cm^2/a)
- M1Max: Upper bound on M1 (in cm^2/a)
- MTracer: Tracer used to estimate mixing intensity
- MComment: Comments on entries about mixing intensity
- L1: Bioturbation depth of the first mixed layer (in cm)
- L2: Bioturbation depth of the second mixed layer (measured from the sediment surface to the depth where mixing drops to 0) (in cm)*
- L1Min: Lower bound on L1 (in cm)
- L1Max: Upper bound on L1 (in cm)
- LTracer: Tracer used to estimate bioturbation depth
- LComment: Comments on the entries about bioturbation depth
- WD: Water depth at the sampling site (in m)
- WDMax: Upper bound on WD (in m)
- WDMIN): Lower bound on WD (in m)
- WDComment: Comment on the entries about water depth
- lat: latitude of the sampling site (in decial degrees)
- lat1, lat2: limits on the latitude of the sampling site (in decial degrees)
- long: longitude of the sampling site (in decial degrees)
- long1, long2: limits on the longitude of the sampling site (in decial degrees)
- coordSource: source of the coordinates of the sampling site
- coordComment: comment on the entries about the sampling site
- coringYear: coring year
- coringMonth: coring month
- coringYearEarliest: earliest possible coring year
- coringMonthEarliest: earliest possible coring month (1-12)
- coringYearLatest: latest possible coring year
- coringMonthLatest: latest possible coring month (1-12)

* below L2, mixing drops to 0

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